

Getting Started

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Instructions

This document runs a simple analysis of the Table 1 from (Wansink and Payne 2009). Edit the header information to show your name and the date you complete the assignment.

Typeset this document in RStudio. You may need to install additional Julia libraries; you can run the code in comments (i.e.)

```
#using Pkg
#Pkg.add("DataFrames")
```

in Julia from the command line to install the necessary libraries. You may also need to run the commented code `install.packages("JuliaCall")` and `julia_setup()` from above in the R console.

If you wish to typeset Julia output to PDF via *LaTeX*, you may need to edit the header to

```
output:
  pdf_document:
    latex_engine: xelatex
```

or some other *LaTeX* command (such as `lualatex`) that is compatible with Unicode.

When you are able to typeset the original document on your computer, modify the code to analyze Calories per Serving. That is, replace all occurrences of the variable names of the form `CaloriesPerServing*` to the variable `CaloriesPerServing*`. Change the values in the table named `ReferenceValues` to match the values in the text of Wansink and Payne (Wansink and Payne 2009).

Typeset the edited document and analyze the results. Add your own comments to the section headed **Comments**. Format your comments as *italicized* text, **bold** text or

as quoted text.

Change the name of this file to match your user name on D2L, keeping the ‘Rmd’ extension, and include week number and coding language in the title (for example, `Peter.Claussen.Julia.1.Rmd`). Upload this file to D2L. Typeset this file to Word or PDF and upload the result to D2L as well.

Data

Table 1: Mean and (SD) for selected recipes from “Joy of Cooking”

Measure	1936	1946	1951	1963	1975	1997	2006
calories per recipe (SD)	2123.8 (1050.0)	2122.3 (1002.3)	2089.9 (1009.6)	2250.0 (1078.6)	2234.2 (1089.2)	2249.6 (1094.8)	3051.9 (1496.2)
calories per serving (SD)	268.1 (124.8)	271.1 (124.2)	280.9 (116.2)	294.7 (117.7)	285.6 (118.3)	288.6 (122.0)	384.4 (168.3)
servings per recipe (SD)	12.9 (13.3)	12.9 (13.3)	13.0 (14.5)	12.7 (14.6)	12.4 (14.3)	12.4 (14.3)	12.7 (13.0)

Analysis

Enter data

```
using Pkg
Pkg.add("DataFrames")
using DataFrames
CookingTooMuch = DataFrame(
  Year=[1936, 1946, 1951, 1963, 1975, 1997, 2006],
  CaloriesPerServingMean1 = [2123.8, 2122.3, 2089.9, 2250.0, 2234.2, 2249.6, 3051.9],
  CaloriesPerServingSD1 = [1050.0, 1002.3, 1009.6, 1078.6, 1089.2, 1094.8, 1496.2],
  CaloriesPerServingMean2 = [268.1, 271.1, 280.9, 294.7, 285.6, 288.6, 384.4],
  CaloriesPerServingSD2 = [124.8, 124.2, 116.2, 117.7, 118.3, 122.0, 168.3],
  ServingsPerRecipeMean = [12.9, 12.9, 13.0, 12.7, 12.4, 12.4, 12.7],
  ServingsPerRecipeSD = [13.3, 13.3, 14.5, 14.6, 14.3, 14.3, 13.0]
);
```

Create values for confidence interval plot

Wansink reports that 18 recipes were analyzed.

```
n = 18;
```

Use standard formula for standard error $\widehat{se} = \hat{\sigma}/\sqrt{n}$ and confidence interval, assuming a significance level α of 5%.

$$CI(95\%) = \hat{\mu} \pm z_{\alpha/2} \times \widehat{se}$$

.

```
Pkg.add("Distributions")
using Distributions

function StandardError(sigma, n)
```

```
sigma/sqrt(n)
end
```

```
## StandardError (generic function with 1 method)
```

```
function ConfidenceBound(sigma, n; alpha = 0.05)
  quantile(Normal(), (1-alpha/2))*StandardError(sigma,n)
end
```

```
## ConfidenceBound (generic function with 1 method)
```

Create a variable for plotting and calculate upper and lower bounds using confidence intervals.

```
PlotCookingTooMuch = CookingTooMuch;
PlotCookingTooMuch[:, :CaloriesPerServing] = PlotCookingTooMuch[:, :CaloriesPerServingMean1];
PlotCookingTooMuch[:, :Lower] = PlotCookingTooMuch[:, :CaloriesPerServing] - ConfidenceBound(PlotCookingTooMuch[:, :CaloriesPerServingSD1], n);
PlotCookingTooMuch[:, :Upper] = PlotCookingTooMuch[:, :CaloriesPerServing] + ConfidenceBound(PlotCookingTooMuch[:, :CaloriesPerServingSD1], n);
PlotCookingTooMuch[:, :Bound] = ConfidenceBound(PlotCookingTooMuch[:, :CaloriesPerServingSD1], n);
PlotCookingTooMuch = PlotCookingTooMuch[:, [:Year, :CaloriesPerServing, :Lower, :Upper, :Bound]];
```

Examine the values to make sure we've entered correctly.

```
println(PlotCookingTooMuch)
```

```
## 7×5 DataFrame
## Row   Year   CaloriesPerServing   Lower   Upper   Bound
##      Int64   Float64           Float64   Float64   Float64
##
##  1   1936           2123.8   1638.73   2181.45   485.066
##  2   1946           2122.3   1659.27   2179.68   463.03
##  3   1951           2089.9   1623.5    2143.58   466.403
##  4   1963           2250.0   1751.72   2304.37   498.279
##  5   1975           2234.2   1731.02   2288.85   503.175
##  6   1997           2249.6   1743.84   2305.96   505.763
##  7   2006           3051.9   2360.7    3129.65   691.196
```

Check our calculations. Wansink reports 95% confidence intervals for the 1936 and 2006 means as [1638.7, 2608.9] and [2360.7, 3743.1], respectively. We should be equal, to within rounding error.

```
CompValues = PlotCookingTooMuch[[1, 7], [:Lower, :Upper]];
ReferenceValues = [[210.4, 325.8] [325.8, 514.7]];
any(broadcast(abs, (Matrix(CompValues) - ReferenceValues)) .> 0.1)
```

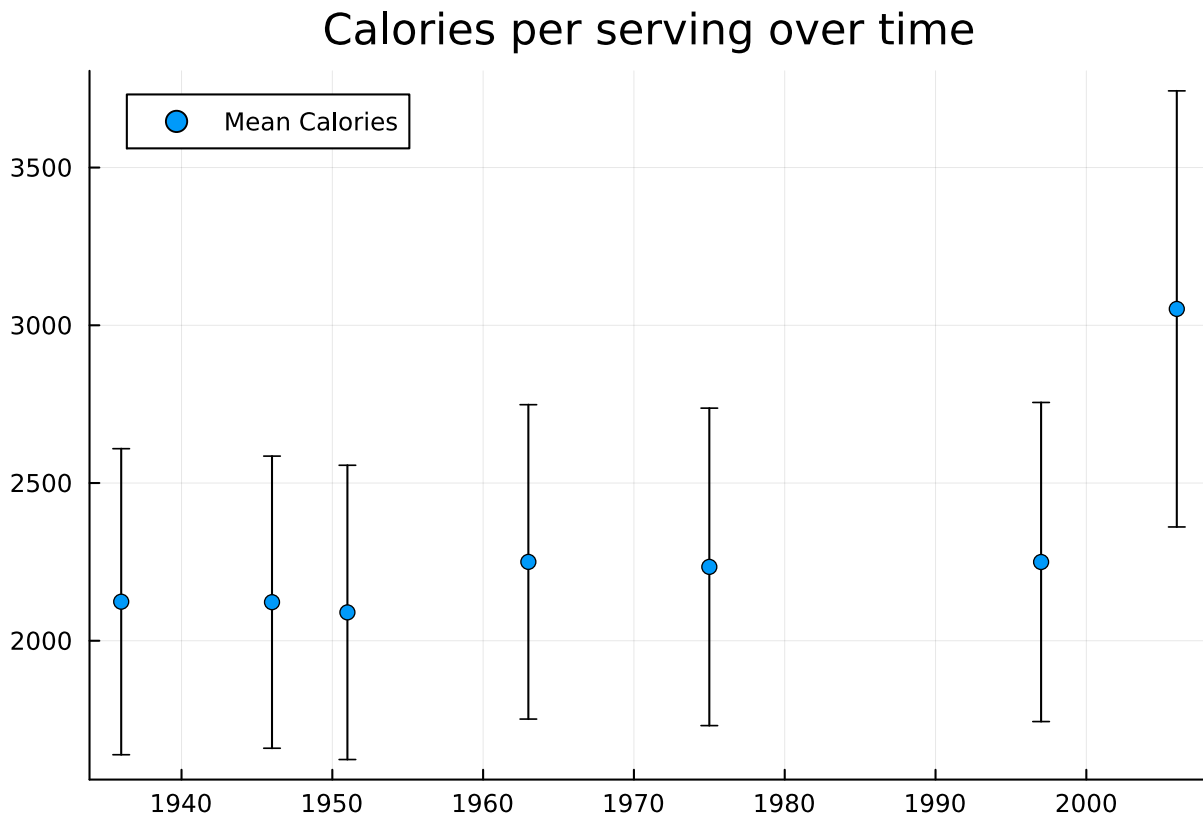
```
## true
```

We no longer need the original data.

```
CookingTooMuch = Nothing;
```

Plot the table

```
Pkg.add("Plots")
Pkg.add("StatsPlots")
using Plots
using StatsPlots
@df PlotCookingTooMuch scatter(:Year, :CaloriesPerServing),
                               yerror=:Bound,
                               title = "Calories per serving over time",
                               labels= "Mean Calories")
```



Comments

From this plot, it appears that average calories per *serving* changed very little from 1936 to 1997. This suggests that the significant increase from 1936 to 2006 reported by Wansink is due to an increase in the 2006 recipes only. Thus, we only need to compare 1936 and 2006 recipe sets.

References

Wansink, Brian, and Collin Payne. 2009. "The Joy of Cooking Too Much: 70 Years of Calorie Increases in Classic Recipes" 150 (150): 291–92.